

- In patients with severe cirrhosis (class C, Child Plugh's Scale), the dosage should be restricted to 25mg/day

Cardiac Impairment

- Baseline ECG is recommended

Elderly

- Baseline ECG is recommended for patients over age 50
- Dose should be reduced to 25 mg/day
- Reduction in the risk of suicidality with antidepressants compared to placebo in adults age 65 and older



Children and Adolescents

- Tianeptine is not recommended for use in children or adolescents under 18
- Carefully weigh the risks and benefits of pharmacological treatment against the risks and benefits of nontreatment with antidepressants and make sure to document this in the patient's chart
- Monitor patients face-to-face regularly, particularly during the first several weeks of treatment
- Use with caution, observing for activation of known or unknown bipolar disorder and/or suicidal ideation, and inform parents or guardians of this risk so they can help observe child or adolescent patients

Pregnancy

- Not recommended for use during pregnancy

Breast Feeding

- Some drug is found in mother's breast milk
- * Not recommended for use during pregnancy
- Immediate postpartum period is a high-risk time for depression, especially in women who have had prior depressive episodes, so drug may need to be reinstituted late in the third trimester or shortly after childbirth to prevent a recurrence during the postpartum period
- Must weigh benefits of breast feeding with risks and benefits of antidepressant treatment versus nontreatment to both the infant and the mother
- For many patients, this may mean continuing treatment during breast feeding

The Art of Psychopharmacology

Potential Advantages

- Elderly patients

Potential Disadvantages

- Patients who have difficulty being compliant with multiple daily dosing

Primary Target Symptoms

- Depressed mood
- Symptoms of anxiety

Pearls

- * Possibly a unique mechanism of action as a glutamatergic antidepressant
- Is not metabolized by CYP450; therefore, the risk of pharmacokinetic drug-drug interactions is minimized

Suggested Reading

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Svenningsson P , Bateup H , Qi H , et al. Involvement of AMPA receptor phosphorylation in antidepressant actions with special reference to tianeptine . *Eur J Neuosci* 2007 ;**26** :3509–17 .
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Topiramate

Therapeutics

Brands

- Topamax
- Epitomax
- Topamac
- Topimax
- Trokendi XR
- Qsymia
- *see index for additional brand names*

Generic?

Yes (not for Qsymia)

Class

- Anticonvulsant, voltage-sensitive sodium channel modulator

Commonly Prescribed for

(bold for FDA approved)

- **Partial onset seizures (for immediate-release: adjunct for adults and pediatric patients 2–16 years of age; for extended-release:**

monotherapy or adjunct for patients 2 years and older)

- **Primary generalized tonic–clonic seizures (adjunct and monotherapy; adults and pediatric patients 2 years of age and older)**
- **Seizures associated with Lennox-Gastaut syndrome (adjunct; 2 years of age and older)**
- **Migraine prophylaxis (ages 12 and older)**
- **Chronic weight management (adjunct to reduced calorie diet and increased physical activity) in adults with an initial body mass index (BMI) of at least 30 kg/m² (obese) or at least 27 kg/m² (overweight) in the presence of at least one weight-related comorbid condition [in combination with phentermine (Qsymia)]**
- **Bipolar disorder (adjunctive; no longer in development)**
- **Psychotropic drug-induced weight gain**
- **Binge eating disorder**



How the Drug Works

- * **Blocks voltage-sensitive sodium channels by an unknown mechanism**
- **Inhibits release of glutamate**
- **Potentiates activity of gamma-aminobutyric acid (GABA)**
- **Carbonic anhydrase inhibitor**

How Long Until It Works

- Should reduce seizures by 2 weeks
- Not clear that it has mood-stabilizing properties, but some bipolar patients may respond and if so, it may take several weeks to months to optimize an effect on mood stabilization

If It Works

- The goal of treatment is complete remission of symptoms (e.g., mania, seizures, migraine)
- Continue treatment until all symptoms are gone or until improvement is stable and then continue treating indefinitely as long as improvement persists
- Continue treatment indefinitely to avoid recurrence of mania, seizures, and headaches

If It Doesn't Work (for Bipolar Disorder)

- * May be effective only in a subset of bipolar patients, in some patients who fail to respond to other mood stabilizers, or it may not work at all
- * Consider increasing dose or switching to another agent with better demonstrated efficacy in bipolar disorder



Best Augmenting Combos for Partial Response or Treatment Resistance

- Topiramate is itself a second-line augmenting agent for numerous other anticonvulsants, lithium, and antipsychotics in treating bipolar disorder

Tests

* Baseline and periodic serum bicarbonate levels to monitor for hyperchloremic, non-anion gap metabolic acidosis (i.e., decreased serum bicarbonate below the normal reference range in the absence of chronic respiratory alkalosis)

Side Effects

How Drug Causes Side Effects

- CNS side effects theoretically due to excessive actions at voltage-sensitive sodium channels
- Weak inhibition of carbonic anhydrase may lead to kidney stones and paresthesias
- Inhibition of carbonic anhydrase may also lead to metabolic acidosis

Notable Side Effects

- * Sedation, asthenia, dizziness, ataxia, parasthesia, nervousness, nystagmus, tremor
- * Nausea, appetite loss, weight loss

- Blurred or double vision, mood problems, problems concentrating, confusion, memory problems, psychomotor retardation, language problems, speech problems, fatigue, taste perversion



Life-Threatening or Dangerous Side Effects

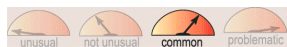
- * Metabolic acidosis
- * Kidney stones
- Hyperammonemia with or without encephalopathy has been reported (may be dose-related; may be more likely with concomitant valproate use)
- Secondary angle-closure glaucoma
- Oligohidrosis and hyperthermia (more common in children)
- Sudden unexplained deaths have occurred in epilepsy (unknown if related to topiramate use)
- Rare activation of suicidal ideation and behavior (suicidality)

Weight Gain



- Reported but not expected
- * Patients may experience weight loss

Sedation



- Many experience and/or can be significant in amount

What to Do About Side Effects

- Wait
- Wait
- Wait
- Take at night to reduce daytime sedation
- Increase fluid intake to reduce the risk of kidney stones
- Switch to another agent

Best Augmenting Agents for Side Effects

- Many side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- Adults, immediate-release: 200–400 mg/day in 2 divided doses for epilepsy; 50–300 mg/day for adjunctive treatment of bipolar disorder
- Adults, extended-release: 200–400 mg/day as adjunct for partial onset seizures; 400 mg/day as adjunct for primary generalized tonic–clonic seizures; 400 mg/day as monotherapy for seizures

- Migraine: initial 25 mg/night for the first week; increase weekly in 25 mg increments; approved dose 100 mg/day in two divided doses

Dosage Forms

- Tablet 25 mg, 50 mg, 100 mg, 200 mg
- Sprinkle capsule 15 mg, 25 mg
- Extended-release capsule 25 mg, 50 mg, 100 mg, 200 mg

How to Dose

- Adults (immediate-release): initial 25–50 mg/day; increase each week by 50 mg/day; administer in 2 divided doses; maximum dose generally 1600 mg/day
- Adults (extended-release, adjunct for seizures): initial 25–50 mg once daily; increase weekly by 25–50 mg
- Seizures (extended-release, monotherapy, patients ages 10 and older): initial 50 mg once daily; increase by 50 mg weekly for 4 weeks, increase by 100 mg weekly for weeks 5 and 6; recommended dose 400 mg/day
- Seizures (immediate-release, ages 2–16): see Children and Adolescents



Dosing Tips

- For migraine, the individual dose may vary widely. Some patients benefit from doses as low as 25 mg/day but others may require

much higher doses than the 100 mg/day approved for migraine prophylaxis

- Headaches may return within days to months of stopping, but patients often continue to do well for 6 or more months after stopping
 - Adverse effects may increase as dose increases
 - Topiramate is available in a sprinkle capsule formulation, which can be swallowed whole or sprinkled over approximately a teaspoon of soft food (e.g., applesauce); the mixture should be consumed immediately
 - Bipolar patients are generally administered doses at the lower end of the dosing range
 - Slow upward titration from doses as low as 25 mg/day can reduce the incidence of unacceptable sedation
 - Many bipolar patients do not tolerate more than 200 mg/day
- * Weight loss is dose-related but most patients treated for weight gain receive doses at the lower end of the dosing range

Overdose

- No fatalities have been reported in monotherapy; convulsions, sedation, speech disturbance, blurred or double vision, metabolic acidosis, impaired coordination, hypotension, abdominal pain, agitation, dizziness

Long-Term Use

- Probably safe
- Periodic monitoring of serum bicarbonate levels may be required

Habit Forming

- No

How to Stop

- Taper
- Epilepsy patients may seize upon withdrawal, especially if withdrawal is abrupt
- * Rapid discontinuation may increase the risk of relapse in bipolar patients
- * Discontinuation symptoms uncommon

Pharmacokinetics

- Elimination half-life approximately 21 hours
- Renally excreted
- Inhibits CYP450 2C19 and induces CYP450 3A4
- Food does not affect absorption

Drug Interactions

- Carbamazepine, phenytoin, and valproate may increase the clearance of topiramate, and thus decrease topiramate levels, possibly requiring a higher dose of topiramate
- Topiramate may increase the clearance of phenytoin and thus decrease phenytoin levels, possibly requiring a higher dose of phenytoin
- Topiramate may increase the clearance of valproate and thus decrease valproate levels, possibly requiring a higher dose of valproate
- Topiramate may increase plasma levels of metformin; also, metformin may reduce clearance of topiramate and increase topiramate levels
- Topiramate may interact with carbonic anhydrase inhibitors to increase the risk of kidney stones
- Topiramate may reduce the effectiveness of oral contraceptives
- Topiramate levels may increase in the presence of hydrochlorothiazide (HCTZ), possibly requiring a dose decrease of topiramate
- Topiramate may decrease the exposure of pioglitazone and its active metabolites; patients taking both medications should be carefully monitored for adequate control of their diabetic disease state
- At high doses, topiramate may increase systemic exposure of lithium

- Reports of hypothermia and hyperammonemia with or without encephalopathy in patients taking topiramate combined with valproate, though this is not due to a pharmacokinetic interaction; in patients who develop unexplained lethargy, vomiting, or change in mental status, an ammonia level should be measured

Other Warnings/Precautions

- * If symptoms of metabolic acidosis develop (hyperventilation, fatigue, anorexia, cardiac arrhythmias, stupor), then dose may need to be reduced or treatment may need to be discontinued
- Depressive effects may be increased by other CNS depressants (alcohol, MAOIs, other anticonvulsants, etc.)
- Use with caution when combining with other drugs that predispose patients to heat-related disorders, including carbonic anhydrase inhibitors and anticholinergics
- Whenever possible, warn patients and their caregivers about the possibility of activating side effects, and advise them to report such side effects immediately

Do Not Use

- Within 6 hours prior to and 6 hours after alcohol use (extended-release)
- In patients with metabolic acidosis who are taking metformin (extended-release)

- If there is a proven allergy to topiramate

Special Populations

Renal Impairment

- Topiramate is renally excreted, so the dose should be lowered by half
- Can be removed by hemodialysis; patients receiving hemodialysis may require supplemental doses of topiramate

Hepatic Impairment

- Drug should be used with caution

Cardiac Impairment

- Drug should be used with caution

Elderly

- Elderly patients may be more susceptible to adverse effects



Children and Adolescents

- Approved for use in children age 2 and older (immediate-release) for treatment of seizures
- Clearance is increased in pediatric patients

- Seizures (immediate-release, ages 2–16): initial 1–3 mg/kg per day at night; after 1 week increase by 1–3 mg/kg per day every 1–2 weeks with total daily dose administered in 2 divided doses; recommended dose generally 5–9 mg/kg per day in 2 divided doses
- Effectiveness was not demonstrated in infants/toddlers 1 to 24 months of age with refractory partial onset seizures; some adverse effects/toxicities not observed in older patients did occur
- Seizures (extended-release, adjunct): initial 25 mg once nightly (1–3 mg/kg/night) for first week; increase at 1- or 2-week intervals by increments of 1–3 mg/kg/night; recommended dose 5–9 mg/kg/night

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Increased risk of cleft lip/palate
- Increased risk of being small for gestational age
- Use in women of childbearing potential requires weighing potential benefits to the mother against the risks to the fetus
- Hypospadias has occurred in some male infants whose mothers took topiramate during pregnancy

- * Lack of convincing efficacy for treatment of bipolar disorder suggests risk/benefit ratio is in favor of discontinuing topiramate in bipolar patients during pregnancy
- * For bipolar patients, topiramate should generally be discontinued before anticipated pregnancies
- Antiepileptic Drug Pregnancy Registry: 1-888-233-2334, <https://www.aedpregnancyregistry.org>
- Taper drug if discontinuing
- * For bipolar patients, given the risk of relapse in the postpartum period, mood stabilizer treatment, especially with agents with better evidence of efficacy than topiramate, should generally be restarted immediately after delivery if patient is unmedicated during pregnancy
- * Atypical antipsychotics may be preferable to topiramate if treatment of bipolar disorder is required during pregnancy
- Bipolar symptoms may recur or worsen during pregnancy and some form of treatment may be necessary
- Seizures, even mild seizures, may cause harm to the embryo/fetus

Breast Feeding

- Some drug is found in mother's breast milk
- * Recommended either to discontinue drug or bottle feed
- If drug is continued while breast feeding, infant should be monitored for possible adverse effects

- If infant shows signs of irritability or sedation, drug may need to be discontinued
- * Bipolar disorder may recur during the postpartum period, particularly if there is a history of prior postpartum episodes of either depression or psychosis
- * Relapse rates may be lower in women who receive prophylactic treatment for postpartum episodes of bipolar disorder
- Atypical antipsychotics and anticonvulsants such as valproate may be safer and more effective than topiramate during the postpartum period when treating nursing mother with bipolar disorder

The Art of Psychopharmacology

Potential Advantages

- Treatment-resistant bipolar disorder
- Patients who wish to avoid weight gain

Potential Disadvantages

- Efficacy in bipolar disorder uncertain
- Patients with a history of kidney stones or risks for metabolic acidosis

Primary Target Symptoms

- Migraine prophylaxis

- Incidence of seizures
- Unstable mood

Pearls

- Side effects may actually occur less often in pediatric patients
 - Has been studied in a wide range of psychiatric disorders, including bipolar disorder, PTSD, binge eating disorder, obesity, and others
 - Some anecdotes, case series, and open-label studies have been published and are widely known suggesting efficacy in bipolar disorder
- * However, randomized clinical trials do not suggest efficacy in bipolar disorder; unfortunately these important studies have not been published by the manufacturer, who has dropped topiramate from further development as a mood stabilizer, though this is not widely known
- * Misperceptions about topiramate's efficacy in bipolar disorder have led to its use in more patients than other agents with proven efficacy, such as lamotrigine
- Migraines may decrease in as little as 2 weeks, but can take up to 3 months on a stable dose to see full effect
 - The goal when using topiramate for migraine prophylaxis is a 50% or greater reduction in migraine frequency or severity; consider tapering or stopping if headaches remit for more than 6 months or if considering pregnancy

- Augmenting options for migraine include beta blockers, antidepressants, natural products, other anticonvulsants, and nonmedication treatments such as biofeedback to improve headache control
- * Due to reported weight loss in some patients in trials with epilepsy, topiramate is commonly used to treat weight gain, especially in patients with psychotropic drug-induced weight gain
- * Weight loss in epilepsy patients is dose-related with more weight loss at high doses (mean 6.5 kg or 7.3% decline) and less weight loss at lower doses (mean 1.6 kg or 2.2% decline)
- * Changes in weight were greatest in epilepsy patients who weighed the most at baseline (>100 kg), with mean loss of 9.6 kg or 8.4% decline, while those weighing <60 kg had only a mean loss of 1.3 kg or 2.5% decline
- * Long-term studies demonstrate that weight losses in epilepsy patients were seen within the first 3 months of treatment and peaked at a mean of 6 kg after 12–18 months of treatment; however, weight tended to return to pretreatment levels after 18 months
- * Some patients with psychotropic drug-induced weight gain may experience significant weight loss (>7% of body weight) with topiramate up to 200 mg/day for 3 months, but this is not typical, is not often sustained, and has not been systemically studied
- Early studies suggest potential efficacy in binge eating disorder

- The combination of topiramate and phentermine is approved as a treatment for obesity (see phentermine/topiramate); the combination is also being studied in diabetes and sleep apnea

Suggested Reading

Johnson BA , Ait-Daoud N . Topiramate in the new generation of drugs: efficacy in the treatment of alcoholic patients . *Curr Pharm Des* 2010 ;**16** (19):2103–12 .

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Tranylcypromine

Therapeutics

Brands

- Parnate
- *see index for additional brand names*

Generic?

Yes

Class

- Neuroscience-based Nomenclature: serotonin, norepinephrine, dopamine multimodal enzyme inhibitor (SN-MM)
- Monoamine oxidase inhibitor (MAOI)

Commonly Prescribed for

(bold for FDA approved)

- **Major depressive episode without melancholia**
- Treatment-resistant depression
- Treatment-resistant panic disorder
- Treatment-resistant social anxiety disorder

How the Drug Works

- Irreversibly blocks monoamine oxidase (MAO) from breaking down norepinephrine, serotonin, and dopamine
 - This boosts noradrenergic, serotonergic, and dopaminergic neurotransmission
- * As the drug is structurally related to amphetamine, it may have some stimulant-like actions due to monoamine release and reuptake inhibition

How Long Until It Works

- Some patients may experience stimulant-like actions early in dosing
- Onset of therapeutic actions usually not immediate, but often delayed 2–4 weeks following adequate dosing
- If it is not working within 6–8 weeks, it may require a dosage increase or it may not work at all
- May continue to work for many years to prevent relapse of symptoms

If It Works

- The goal of treatment is complete remission of current symptoms as well as prevention of future relapses
- Treatment most often reduces or even eliminates symptoms, but not a cure since symptoms can recur after medicine stopped
- Continue treatment until all symptoms are gone (remission)
- Once symptoms are gone, continue treating for 1 year for the first episode of depression

- For second and subsequent episodes of depression, treatment may need to be indefinite
- Use in anxiety disorders may also need to be indefinite

If It Doesn't Work

- Many patients have only a partial response where some symptoms are improved but others persist (especially insomnia, fatigue, and problems concentrating)
- Other patients may be nonresponders, sometimes called treatment-resistant or treatment-refractory
- Some patients who have an initial response may relapse even though they continue treatment, sometimes called “poop-out”
- Consider increasing dose, switching to another agent, or adding an appropriate augmenting agent
- Consider psychotherapy
- Consider evaluation for another diagnosis or for a comorbid condition (e.g., medical illness, substance abuse, etc.)
- Some patients may experience apparent lack of consistent efficacy due to activation of latent or underlying bipolar disorder, and require antidepressant discontinuation and a switch to a mood stabilizer



Best Augmenting Combos for Partial Response or Treatment Resistance

* Augmentation of MAOIs has not been systematically studied, and this is something for the expert, to be done with caution and with careful